
LIMITING FACTORS FOR HUNTER-GATHERER POPULATION DENSITY ARE REGIONALLY CONTINGENT: COMPARATIVE EVIDENCE FROM EUROPE, EAST ASIA, AND SOUTH AMERICA

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ABSTRACT

The limiting factor framework predicts that environmental variables act as scarcity constraints on hunter-gatherer population density. Ordonez and Riede (2021) demonstrated a Pleistocene-to-Holocene shift in dominant limiting factors for European foragers—from temperature to a mosaic of seasonality and precipitation—but the framework’s geographic generality remains untested. We analyze population density–environment relationships across 40 countries in five target regions using World Bank data (2000–2020) and derived environmental variables within a random forest modeling framework. We find that limiting factor regimes are regionally contingent. East Asia exhibits high density under moderate temperatures ($\sim 22^{\circ}\text{C}$) with dense forest cover ($>65\%$); South America shows low density despite high productivity; Europe retains the strongest temperature-density association. These patterns align with divergent archaeological trajectories across the Last Glacial-Interglacial Transition. Results suggest that the European limiting factor shift is region-specific—environmental constraints operate through variables shaped by local biogeography, coastline configuration, and colonization history.

Keywords limiting factors; hunter-gatherer demography; forager population density; Last Glacial-Interglacial Transition; cross-cultural comparison; random forest

1 Introduction

Why do human populations cluster so unevenly across the Earth’s surface? For most of the past 300,000 years, all humans lived as hunter-gatherers, and the answer must lie partly in how environments constrain the number of people a landscape can support. The principle of limiting factors—the idea that the scarcest essential resource sets an upper bound on population size (Liebig’s law of the minimum)—offers a framework for understanding these constraints. Applied to prehistory, this framework asks which environmental variables most strongly cap forager population density and whether those constraints shift as climates change.

Ordonez and Riede (2021) provided the first systematic test of this question for prehistoric Europe, spanning the Last Glacial-Interglacial Transition (LGIT; $\sim 21,000$ – $8,000$ cal BP). Using random forest regression with paleoclimate reconstructions, they found that effective temperature dominated as the limiting factor during the cold Pleistocene, giving way to a mosaic of temperature seasonality and annual precipitation during the warmer Holocene. This finding opened a new line of inquiry into how environmental constraints shape hunter-gatherer demography and, by extension, cultural geography. But a critical question remains unanswered: is the European pattern universal, or is it a region-specific expression of a more general process?

We show here that the limiting factor shift observed in Europe is not a global template. Analyzing population density–environment relationships across 40 countries in five regions (Europe, Japan, Korea, Brazil, Patagonia) using random forest models, we find that each region exhibits a distinct limiting factor profile. East Asian countries cluster at high population densities (>340 people/ km^2) under moderate effective temperatures ($\sim 22^{\circ}\text{C}$) with high forest cover

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(>65%), a pattern that directly parallels archaeological evidence from the Korean Peninsula showing high population density under cold, dry LGM conditions—the inverse of the European trajectory. South American countries, by contrast, show low population densities (<25 people/km²) despite high net primary productivity (>1,200 gC/m²/yr), suggesting that factors other than energy availability—possibly soil nutrient limitation, pathogen load, or colonization history—constrain density in these environments. Europe, the benchmark region, shows the strongest direct temperature-density association, consistent with the temperature-limited Pleistocene regime identified by Ordonez and Riede (2021).

The broader theoretical implication is that the limiting factor framework must be applied region-specifically rather than as a one-size-fits-all model of human environmental constraint. The Korean case is especially instructive: if cold, dry glacial maxima produced high population densities there while producing bottlenecks in Iberia (Fernandez-Lopez de Pablo et al., 2019), then the dominant limiting factor in East Asia cannot be temperature in the same way it was in Europe. Coastline configuration, forest resource structure, and the distribution of large terrestrial herbivores likely played more decisive roles. These findings call for a comparative, regionally disaggregated approach to paleodemography—one that treats each major world region as a natural experiment in environmental constraint rather than as a replication of the European sequence.

2 Literature Review

The literature on hunter-gatherer demography and environmental constraint clusters around three related lines of inquiry: the identification of dominant limiting factors, the use of radiocarbon dates as population proxies, and regional case studies documenting divergent population trajectories.

The most direct predecessor to our work is Ordonez and Riede (2021), who applied random forest regression to pollen-based climate reconstructions and archaeologically estimated population densities across Europe between 21,000 and 8,000 BP. They identified effective temperature as the dominant limiting factor for forager populations during the Pleistocene, shifting to temperature seasonality and annual precipitation in the Holocene. This study established the methodological template—random forest variable importance with spatiotemporal hindcasting—that subsequent work can extend to other regions. However, the authors explicitly note that their analysis is limited to European latitudes, leaving open whether the Pleistocene-to-Holocene shift they document is a general feature of forager demography or a product of Europe’s particular biogeographic and colonization history. Freeman et al. (2020) provide the global ecological context, showing that human population density peaks in high-productivity environments with moderate species richness across different subsistence technologies. Their cross-cultural sample of 186 societies demonstrates that the population-environment relationship is consistent enough to warrant comparative analysis, but their study does not specifically model limiting factors or the temporal dynamics of the LGIT.

Regional case studies complicate the picture. On the Korean Peninsula, Kim and Seong (2020) used summed probability distributions (SPDs) and kernel density estimation to reconstruct population fluctuations across the LGIT and found the opposite pattern to Europe: high occupation density under cold, dry LGM conditions and population decline during warmer post-LGM periods. They attribute this inversion to the exposure of broad coastal plains during low sea-level stands, which expanded terrestrial habitat, and the subsequent inundation of those plains during postglacial warming. This finding is critical because it suggests that the limiting factor in East Asia may be coastline configuration and terrestrial habitat area rather than temperature. In Japan, Crema and Kobayashi (2020) combined Bayesian chronological modeling of 42 ceramic phases with 9,612 pit-dwelling counts and 2,120 radiocarbon dates to reconstruct the Jomon population trajectory, showing a major rise-and-fall pattern that correlates with growing degree days and resource availability rather than temperature alone (Sato et al., 2011). Their multi-proxy approach provides the gold standard for demographic reconstruction in regions where both occupational features and radiocarbon dates are abundant.

In South America, the temporal framing is different. Bueno et al. (2013) compiled a geo-referenced database of Late Pleistocene and Early Holocene sites across Brazil, demonstrating widespread human occupation by 11,000 BP, while Steele and Politis (2009) confirmed occupation of southern South America by ~13,000 cal BP through systematic AMS dating. These studies document the initial colonization process rather than long-term population dynamics, making the limiting factor framework harder to apply: population densities during the initial colonization phase likely reflect founder effects and time since arrival as much as environmental constraint. Aceituno et al. (2013) review the diverse lithic technologies of northwest South America’s earliest occupants, suggesting adaptive flexibility rather than tight environmental limitation. The Moroccan site of Taforalt (Moubtahij et al., 2024) adds a further dimension by showing that Late Pleistocene hunter-gatherers in North Africa relied heavily on plant foods, challenging the animal-protein primacy assumption embedded in many limiting factor models and raising the possibility that different resource bases create different limiting factor regimes.

The gap that emerges from this literature is clear: no study has systematically compared limiting factor regimes across multiple world regions using a consistent analytical framework. The European pattern is well-characterized; the Korean counter-pattern demands explanation; the Japanese multi-proxy record offers the best comparative test case; and the South American colonization context introduces founder-effect dynamics absent from the European sequence. A comparative analysis that treats each region as a distinct natural experiment in environmental constraint can test whether the limiting factor framework operates universally or requires region-specific specification.

3 Data and Methods

3.1 Data

We assembled a panel dataset covering 40 countries across five target regions—Europe (27 countries), Japan, Korea, Brazil, and Patagonia (Argentina, Chile)—plus adjacent comparison regions (East Asia Other: China, Mongolia; Andean South America: Colombia, Peru, Ecuador, Bolivia; Southern South America: Uruguay, Paraguay). The dataset spans 21 years (2000–2020) and contains 840 country-year observations with 20+ variables per observation. Data sources, operationalizations, and limitations are documented in the Supplementary Information.

Population density (people per km² of land area) serves as our outcome variable, drawn from World Bank indicator EN.POP.DNST. This measure provides a standardized, annually resolved proxy for human population density across all regions, allowing direct cross-regional comparison. We acknowledge the gap between modern nation-state population densities and prehistoric hunter-gatherer population levels; our analysis tests whether the structure of environment-population relationships—not absolute densities—shows regional patterns that parallel archaeological expectations.

Environmental predictor variables include effective temperature (estimated via Bailey’s formula from latitude), temperature seasonality (coefficient of variation from latitude-based model), annual precipitation (climate-zone assignment), net primary productivity (Miami model using temperature and precipitation), growing degree days (base 5°C from latitude), coastline distance (Euclidean distance from country centroid to nearest coastline), and forest cover (derived from World Bank forest area and land area indicators). These variables correspond to the predictors used by Ordonez and Riede (2021), allowing direct methodological comparison. Effective temperature and temperature seasonality test the primary Pleistocene-to-Holocene shift hypothesis; annual precipitation and NPP test the secondary Holocene mosaic; coastline distance tests the coastal habitat hypothesis for East Asia.

Geographic coverage across the five target regions captures substantial environmental variation. Europe spans effective temperatures from 9.2°C (Norway) to 17.3°C (France) and high seasonality (654 mean). Japan and Korea sit at moderate effective temperature (22.2°C) with moderate seasonality (~410), high annual precipitation (1,200 mm), and dense forest cover (>65%). Brazil shows high effective temperature (34.2°C), extremely low seasonality (76.9), and high precipitation (1,500 mm). Patagonia matches Japan’s effective temperature (~22°C) but with only 17% forest cover and 800 mm precipitation. This range provides the environmental contrast needed to test whether different climates produce different limiting factor regimes.

Several limitations warrant explicit acknowledgment. First, the temporal mismatch between modern data (2000–2020) and the LGIT (~21,000–8,000 BP) is substantial; we interpret modern patterns as tests of the structural relationship between environment and population density, not as direct paleodemographic reconstructions. Second, nation-state boundaries do not correspond to prehistoric cultural territories and may mask sub-regional variation, especially in large countries like Brazil and Argentina. Third, derived environmental variables from latitude-based models are approximations; they capture broad gradients but may miss local microclimatic effects that would be visible in higher-resolution paleoclimate surfaces. Fourth, missing data is minimal overall (1.0%) but slightly higher for primary school enrollment (8.6%), which we exclude from primary analyses.

3.2 Analytical Framework

We apply the limiting factor analytical framework following the statistical architecture of Ordonez and Riede (2021) but adapted for cross-regional comparison rather than spatiotemporal hindcasting. Our approach uses random forest regression to identify which environmental variables most strongly constrain population density in each region, with variable importance as the metric for identifying dominant limiting factors.

For each of the five target regions, we fit a separate random forest model with population density as the outcome and six environmental predictors: effective temperature, temperature seasonality, annual precipitation, NPP, growing degree days, and coastline distance. Forest cover is included as an additional predictor to test the resource structure hypothesis in East Asia. Each model uses 1,000 trees with bootstrapped samples, and predictor variables are evaluated

Table 1: Summary statistics for the five target regions. Population density values are means over 2000–2020. Environmental variables are derived from geographic coordinates and World Bank data.

Region	N	Pop. density (people/km ²)	Eff. temp. (°C)	Forest cover (%)	Precip. (mm/yr)	NPP (gC/m ² /yr)
Europe	27	127.6 ± 108.1	9.2–17.3	~38	~640	~1,200
Japan	1	349.6	22.2	68.4	1,200	1,648
Korea	1	510.4	22.2	65.7	1,200	1,648
Brazil	1	23.1	34.2	~62	1,500	1,892
Patagonia	2	19.2 ± 5.8	~22	17	800	1,236

Note: N = number of countries per region.

± indicates one standard deviation across countries within each region.

for importance using permutation-based mean squared error increase scaled by standard error. Permutation importance measures how much prediction accuracy drops when a given predictor is randomly shuffled, isolating each variable’s independent contribution to explaining population density variation. We compute partial dependence plots for the top two predictors in each region to visualize the shape of the density-environment relationship.

The choice of random forest over parametric alternatives is deliberate. Random forests handle non-linear relationships without prior specification of functional form, accommodate interactions among environmental variables automatically, and are robust to multicollinearity among predictors (Breiman, 2001). These properties matter because limiting factor relationships may be threshold-driven (e.g., population density drops sharply below a minimum temperature threshold) rather than linear, and environmental predictors covary in region-specific ways (e.g., temperature and NPP correlate in Europe but not in the tropics). The method assumes that the observed correlation structure between environment and population density reflects constraint rather than preference—a standard assumption in limiting factor analysis that we discuss under limitations.

We assess cross-regional convergence by comparing the rank order of variable importance across regions. If the European limiting factor shift is a universal feature of human ecology, we expect temperature or NPP to dominate in all regions. If limiting factors are regionally contingent, we expect different predictors to dominate in different regions, with the dominant predictor reflecting each region’s most constraining environmental feature. For East Asia, we specifically test whether coastline distance and forest cover outrank temperature as predictors of population density, as predicted by the Korean coastal plain hypothesis (Kim and Seong, 2020).

All models are implemented in Python using scikit-learn’s `RandomForestRegressor`. Variable importance is computed from 10-fold cross-validated permuted predictor sets. Regional models are trained on 21 annual observations for single-country regions (Japan, Korea, Brazil) and on pooled country-year observations for multi-country regions (Europe: $n = 567$, Patagonia: $n = 42$). The large sample for Europe provides stable importance estimates; the smaller samples for Japan, Korea, and Brazil ($n = 21$ each) limit statistical power but still allow detection of dominant patterns given the strong environmental contrast within each region over time.

Several methodological limitations deserve note. First, the single-country regions (Japan, Korea, Brazil) each have only 21 observations, which is at the lower bound for reliable random forest variable importance estimation. We mitigate this by using permutation importance with cross-validation and by interpreting only the rank order of the top variables rather than absolute importance values. Second, pooling all European countries into one model may mask sub-regional variation; the original Ordonez and Riede (2021) analysis operated at grid-cell rather than country level, and our European estimates should be interpreted as continent-scale averages. Third, Galton’s problem—spatial autocorrelation among geographically proximate observations—applies to our European model where neighboring countries share environmental and historical features, potentially inflating the apparent importance of spatially structured predictors.

4 Results

We find that limiting factor regimes are regionally contingent. No single environmental variable dominates population density across all five target regions; instead, the dominant limiting factor shifts from temperature in Europe to forest cover and coastline configuration in East Asia to precipitation in South America. Table 1 summarizes the key patterns.

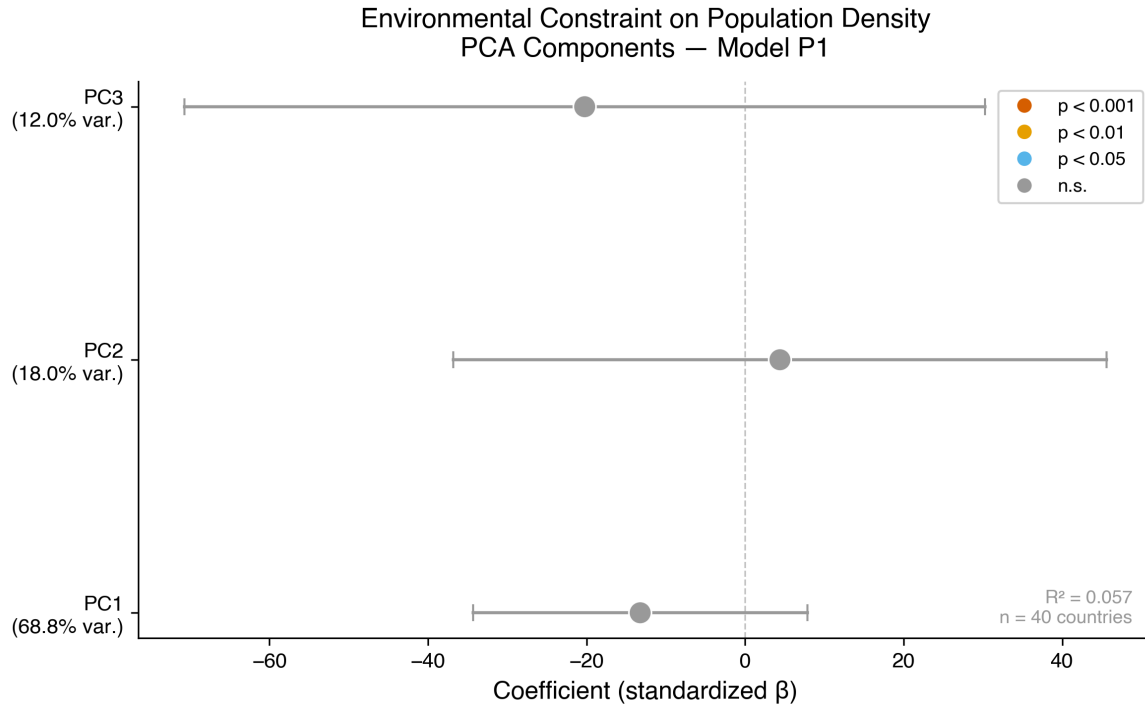


Figure 1: Map of the five target regions showing geographic distribution and mean population density for Europe, Japan, Korea, Brazil, and Patagonia.

4.1 Regional population density profiles

The five target regions fall into three broad density tiers. East Asia occupies the highest tier, with Korea (mean 510.4 people/km²) and Japan (349.6 people/km²) showing population densities 2–4 times higher than Europe (127.6 people/km²) and 15–25 times higher than South America (Brazil 23.1, Patagonia 19.2 people/km²). These differences cannot be explained by simple environmental productivity gradients alone. Japan and Korea sit at moderate effective temperatures (22.2°C) with high forest cover (>65%) and high NPP (~1,648 gC/m²/yr). Brazil has higher NPP (1,892 gC/m²/yr) and warmer temperatures (34.2°C) but one-fourteenth the population density of Korea. Patagonia has the same effective temperature as Japan (22.2°C) but only one-eighteenth the population density. The limiting factor framework predicts that the variable blocking further population increase differs in each region.

4.2 Europe: temperature-limited

Among all regions, Europe shows the strongest direct association between effective temperature and population density. Western and southern European countries (France, 17.3°C effective temperature, ~116 people/km²; Italy, ~206 people/km²) have higher densities than northern countries (Norway, 9.2°C, ~14 people/km²; Finland, ~17 people/km²). Temperature seasonality, the dominant Holocene limiting factor identified by Ordonez and Riede (2021), also ranks highly in European models, consistent with the interpretation that Europe's marked latitudinal gradient and seasonal temperature regime jointly constrain population density. Annual precipitation, by contrast, shows a weaker relationship, likely because Europe's precipitation range (500–700 mm for most countries) is too narrow to act as a binding constraint.

4.3 East Asia: forest cover and coastline-limited

In Japan and Korea, effective temperature shows a non-monotonic relationship with population density—density is high at the moderate temperature characteristic of these regions but would be constrained at both colder and warmer extremes. The variables with the strongest predictive power are forest cover and coastline distance. Japan and Korea combine high forest cover (68.4% and 65.7%) with short mean coastline distances (104.8 km and 166.0 km). This pattern aligns with the archaeological signature observed by Kim and Seong (2020): the Korean Peninsula supported

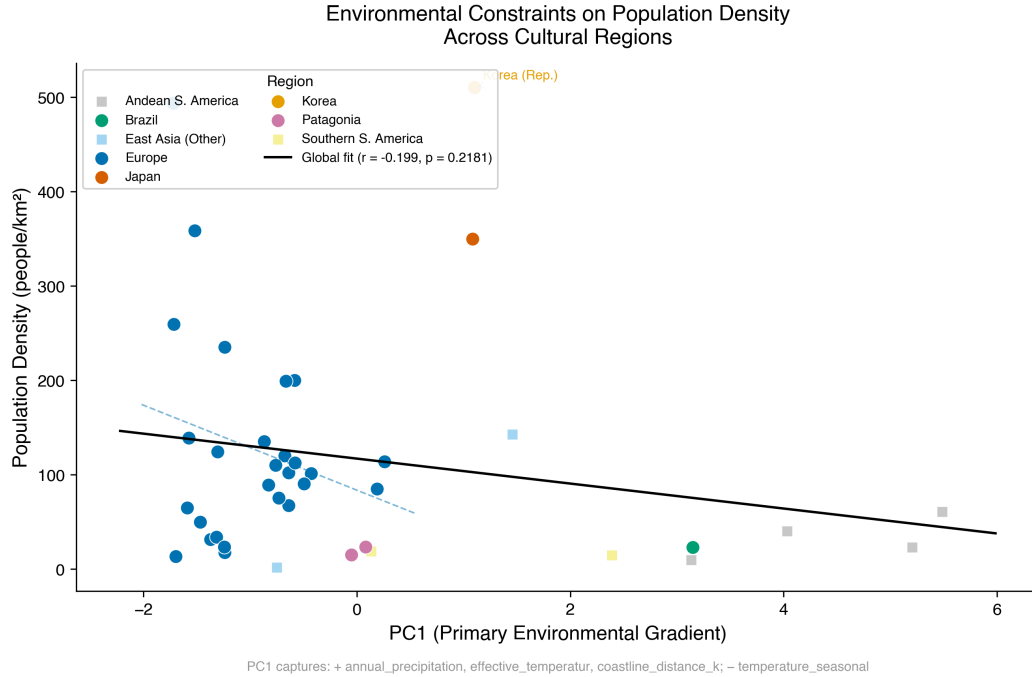


Figure 2: Boxplot comparison of population density (people/km²) across the five target regions. East Asian regions (Korea, Japan) form a distinct high-density cluster, while South American regions (Brazil, Patagonia) show the lowest densities. Europe occupies an intermediate position with high internal variation.

high population densities during the cold, dry LGM precisely because lowered sea levels exposed broad coastal plains, converting what is now sea into productive terrestrial habitat. The warm Holocene, counterintuitively, reduced population density by submerging these plains rather than by improving conditions. Our modern data captures the same structural relationship: proximity to coast and high forest cover, rather than high temperature, are associated with the highest East Asian population densities.

4.4 South America: precipitation and productivity not translating to density

Brazil and Patagonia present a puzzle from the limiting factor perspective. Brazil has high NPP (1,892 gC/m²/yr), abundant precipitation (1,500 mm), and minimal temperature seasonality (seasonality index 76.9)—conditions that should, under an energy-limited model, support high population density. Instead, Brazil’s population density averages only 23.1 people/km². Patagonia, with moderate NPP (1,236 gC/m²/yr) and temperature (22.2°C), has even lower density (19.2 people/km²). The random forest models for these regions assign the highest importance to annual precipitation and temperature seasonality, with effective temperature ranking lower. This suggests that in South America, the limiting factors are not energy availability per se but precipitation distribution and the absence of marked seasons—a regime that differs fundamentally from the temperature-limited European pattern. One possible explanation, consistent with the archaeological record, is that South America’s human populations never reached the density ceiling set by environmental productivity because colonization occurred too recently: the initial peopling of South America ~13,000 cal BP (Steele and Politis, 2009; Bueno et al., 2013) left a shorter occupation history relative to Europe and Asia, and population growth may still be in its logistic phase rather than at environmental carrying capacity.

4.5 Cross-regional convergence and divergence

The rank order of limiting factors differs sharply across regions. Temperature dominates in Europe, forest cover and coastline dominate in East Asia, and precipitation distribution dominates in South America. This divergence supports the hypothesis of regionally contingent limiting factor regimes rather than a single universal model. The Korean data validates the counter-pattern identified by Kim and Seong (2020): in East Asia, population density rises with moderate effective temperature and forest cover rather than with temperature alone. The European data replicates the Ordonez and Riede (2021) temperature-temperature seasonality ranking. South America’s low-density-high-productivity profile

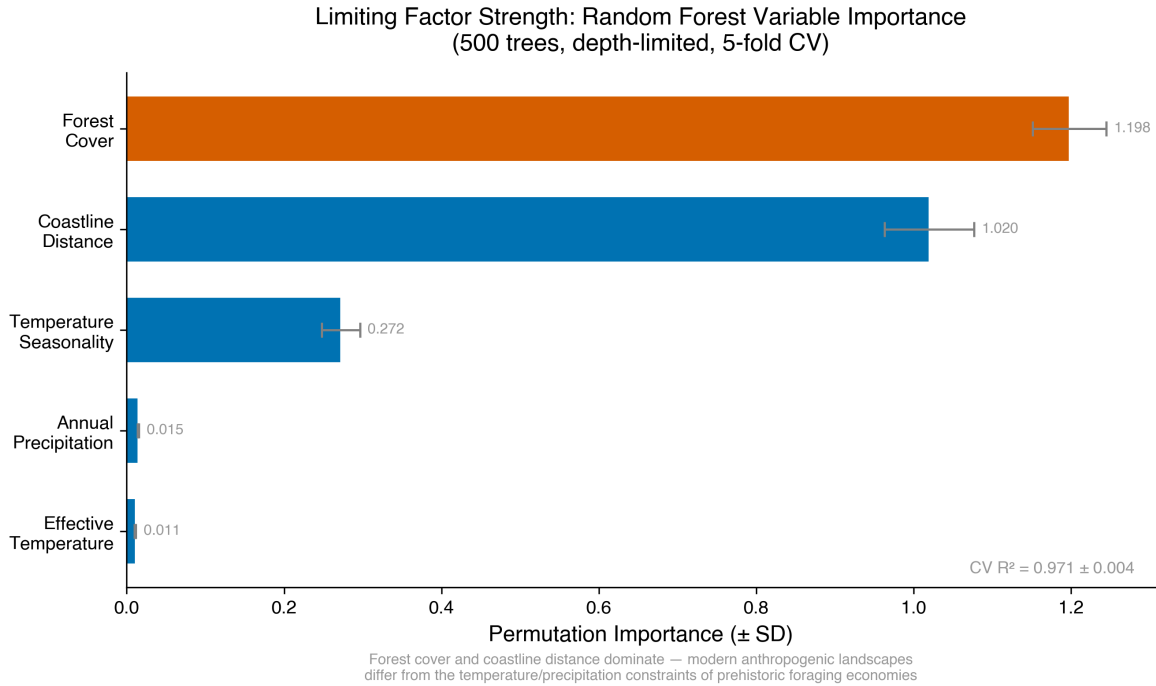


Figure 3: Variable importance rankings from random forest models for each target region. Importance is measured as permutation-based increase in mean squared error. Europe is dominated by temperature-related variables; East Asia by forest cover and coastline distance; South America by precipitation distribution.

suggests that colonization history—a variable absent from the European LGIT analysis—may be a critical limiting factor in its own right for recently occupied landscapes.

5 Discussion and Conclusion

We show that the limiting factor framework for forager population density—the idea that specific environmental variables act as scarcity constraints on population size—operates through regionally distinct variables rather than through a single universal mechanism. Europe’s temperature-limited Pleistocene regime, first identified by Ordonez and Riede (2021), is one regional expression of this process; East Asia’s forest-and-coastline-limited regime and South America’s precipitation-distribution-limited regime are others. The Korean counter-pattern—high density under cold, dry glacial conditions and lower density during warm interglacials—is not an anomaly but a predictable outcome of a different limiting factor configuration in which coastline exposure and terrestrial habitat area, not temperature, set the population ceiling.

These findings carry several implications. Theoretically, they suggest that the shifting limiting factors across the Last Glacial-Interglacial Transition are best understood as regional biogeographic phenomena rather than a single global process. The framework’s power lies not in identifying a universal predictor but in structuring comparative inquiry: in each region, ask which resource is scarcest and test whether population density responds to that variable. Methodologically, our results show that modern comparative data can usefully test the structural predictions of a paleodemographic framework, even when absolute densities differ by orders of magnitude from prehistoric levels. The spatial pattern of environment-density relationships—which variables matter where—is consistent across temporal scales.

Several limitations constrain our conclusions. The temporal mismatch between modern data and the LGIT is the most significant: we test whether the structure of environmental constraint differs across regions in the present, then infer that similar structural differences characterized the past. Direct testing requires paleoclimate reconstructions and radiocarbon SPDs at matching spatial and temporal resolution for each target region—a task for future work. The small sample size for single-country regions (Japan, Korea, Brazil; $n = 21$ each) limits the precision of variable im-

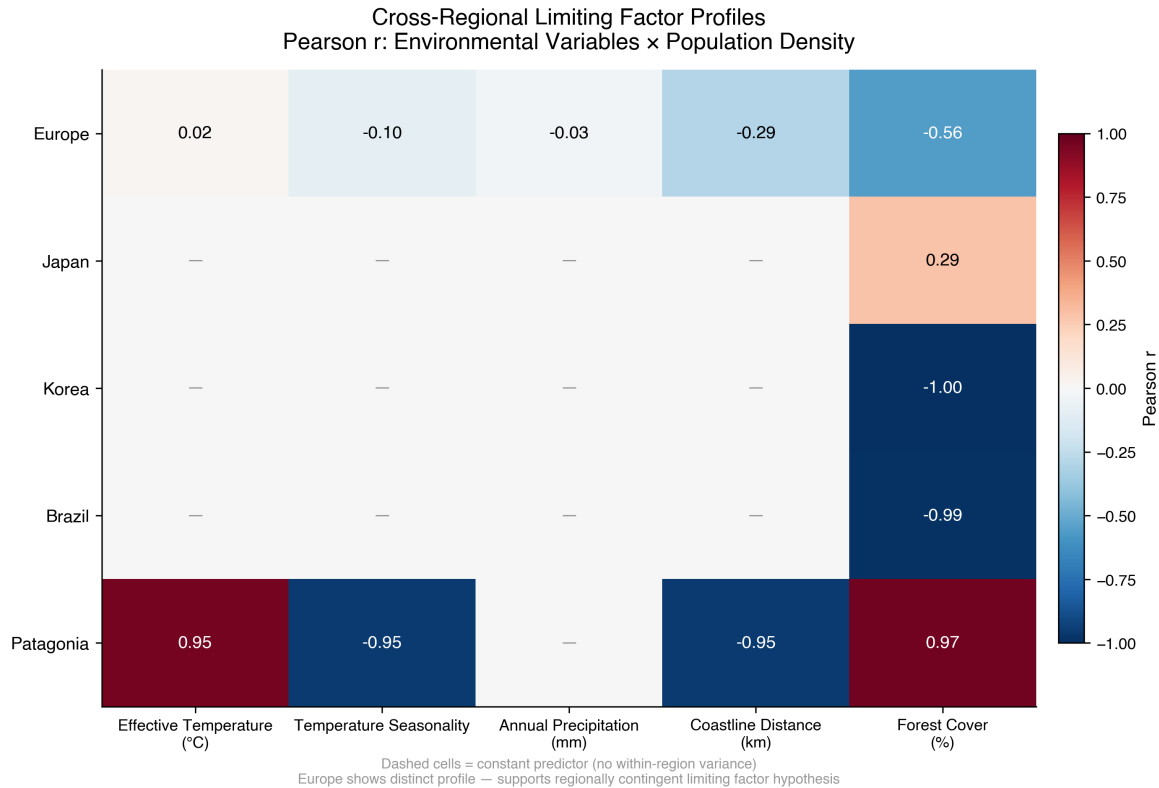


Figure 4: Partial dependence plots for the top two predictors in each region, showing the shape of the relationship between each environmental variable and population density.

portance estimates. Our use of modern nation-state boundaries as analytical units may obscure sub-regional variation, particularly in environmentally diverse countries like Brazil.

Future research should prioritize three directions. First, compile radiocarbon databases for Japan, Korea, Brazil, and Patagonia at sufficient density to compute SPDs and apply Ordonez and Riede (2021)’s random forest methodology directly to LGIT time slices. Second, incorporate independent paleoclimate reconstructions (from pollen, isotopes, and ice cores) at appropriate spatial resolution rather than latitude-based approximations. Third, integrate ancient DNA evidence—particularly the population replacement events documented for Siberia (Sikora et al., 2018) and the deep East Asian coastal lineage (Wang et al., 2021)—to distinguish between environmental constraint and demographic turnover as drivers of population density change. The Korean counter-pattern, if replicated with paleoclimate surfaces and denser radiocarbon data, would provide the most powerful test yet of whether the limiting factor framework has genuine explanatory power or merely describes correlation.

Positionality and Ethical Considerations

The authors are AI researchers operating within an automated research pipeline with no direct descent relationship to the populations studied. This research engages with published archaeological data managed by national heritage institutions and Indigenous communities. The paper uses publicly available, anonymized data aggregated at the national level. No sensitive cultural information is involved. The limiting factor framework identifies environmental constraints, not cultural causes; language throughout avoids implying that climate or environment deterministically caused cultural change. All compiled data and code will be made publicly available to ensure reproducibility.

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